

FOOD ABUNDANCE, INTRUDER PRESSURE, AND BODY SIZE AS DETERMINANTS OF TERRITORY SIZE IN JUVENILE STEELHEAD TROUT (*ONCORHYNCHUS MYKISS*)

by

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Summary

We observed the behaviour of juvenile steelhead trout (*Oncorhynchus mykiss*) in two rivers in British Columbia, Canada, to determine the importance of invertebrate drift abundance, intruder pressure, and body size of territory holders as correlates of territory size. Territory size increased with increasing fish size, but fish smaller than 5 cm had relatively large territories for their body size, in comparison to fish that were larger than 5 cm in length. After statistically controlling for the effect of body size, territory size was inversely related to the abundance of drifting invertebrates. Territory size was not related to the number of intrusions observed on a territory per hour, but was inversely related to local fish density, a second measure of intruder pressure. Steelhead trout appear to defend territory areas that are most similar in size to juvenile Atlantic salmon. This study provides evidence that stream-dwelling salmonids scale territory size to body size in a manner that allows them to capture a maximum daily ration of food on their territories.

Introduction

The study of territory size has been approached by ecologists in two distinctly different ways. An optimality approach predicts how large an area

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an animal should defend based on the costs and benefits of defence that are largely determined by environmental conditions (Davies & Houston, 1984). Empirical tests of this approach have usually shown that territory size is inversely related to food abundance (Stimson, 1973; Simon, 1975; Hixon, 1980) or intruder pressure (Myers *et al.*, 1979; Norman & Jones, 1984; Eberhard & Ewald, 1994). A second way ecologists have approached the study of territory size has been through the use of interspecific allometric regressions. Allometric studies have generally found that territory or home range size increases with body mass of a species and decreases with the amount of plant matter in the diet (Schoener, 1968; Harestad & Bunnell, 1979). These studies are almost always applied interspecifically probably because most species included in allometric studies of space requirements do not defend territories over a wide range of body sizes (but see Hart, 1985). In contrast, optimal territory size models are usually tested intraspecifically, but again, ignore the variation in body size found within a species.

In certain groups of animals, such as insects (Hart, 1985), fish (Elliott, 1990), lizards (Simon, 1975), molluscs (Stimson, 1973) and crustaceans (Wada, 1993), territory defence can occur over a wide range of environmental conditions and body sizes. Thus, a combination of both an optimality and allometric approach seems appropriate for studying territory size in such species. Stream-dwelling salmonid fishes are good candidates for a study that combines both approaches. Salmonid fishes defend feeding territories that increase in size with body size (Elliott, 1990; Grant & Kramer, 1990; Keeley & Grant, 1995) and are influenced by environmental factors such as food abundance (Slaney & Northcote, 1974; Dill *et al.*, 1981; Keeley & Grant, 1995) and intruder pressure (Dill *et al.*, 1981; McNicol & Noakes, 1984). Taken together, allometric and environmental correlates of territory size in salmonid fishes suggests that territory size is strongly influenced by body size (Elliott, 1990; Grant & Kramer, 1990; Keeley & Grant, 1995) and only weakly related to environmental factors such as food abundance and intruder pressure (Dill *et al.*, 1981; Keeley & Grant, 1995). This suggests that for a given body size, territory size is relatively fixed and therefore could limit local fish density; however, no study, has attempted to combine both an allometric and optimality approach to predict territory

size over a range of environmental conditions that is comparable to the range of body sizes a salmonid fish will exhibit while defending a territory.

In this paper, we present the results of a field study on juvenile steelhead trout (*Oncorhynchus mykiss*) that was designed to determine the relative importance of body size, food abundance, and intruder pressure as determinants of territory size. To compare the aggressive behaviour of individuals under extreme ranges of body sizes and stream conditions, we observed territorial individuals either in a highly productive stream or in a stream of lower productivity, within the same watershed. We tested the predictions that territory size increases with body size (Elliott, 1990; Grant & Kramer, 1990) and is inversely related to food abundance and intruder pressure (Dill *et al.*, 1981; McNicol & Noakes, 1984; Keeley & Grant, 1995).

Methods

Study areas and species

To maximize the range of environmental conditions, we observed territorial steelhead trout fish in two streams, the Chilliwack and Salmon rivers. Both are tributaries to the Fraser River in southwestern British Columbia, Canada. The Chilliwack River (49°05.5'N, 121°36.0'W) is a relatively steep gradient stream originating in the Cascade Mountains in northern Washington State. The Salmon River (49°08.5'N, 122°36.0'W) is a relatively low gradient stream, originating in wooded farmland, northeast of Langley, British Columbia. In comparison to the upper reaches of the Chilliwack River, the Salmon River has a much higher productivity level for periphyton, invertebrates, and fish (Hartman, 1965, 1968; E.A. Parkinson, pers. comm., B.C. Ministry of Environment, Vancouver, B.C.). Both streams support naturally reproducing populations of steelhead trout, the anadromous form of the rainbow trout (Scott & Crossman, 1973). Adult steelhead trout mature in the ocean and return to spawn in the streams of southwestern British Columbia from mid-April to mid-May each spring (Hartman, 1965, 1968; Ward & Slaney, 1988). Young-of-the-year steelhead trout emerge from gravel nests at about 3 cm in length by mid-June and remain in the streams for 2-3 years before migrating to the ocean at about 17 cm in length (Maher & Larkin, 1954). Within streams, juvenile steelhead trout maintain a foraging station by swimming against the current and darting out to capture invertebrate prey drifting in the water column. In addition to feeding from these stations, juvenile steelhead trout also aggressively defend these sites by chasing intruding fish from around their foraging area (Hartman, 1965). Although 13 species of fish occur in the Chilliwack River and 16 species occur in the Salmon River (Hartman, 1968), in the areas examined in this study, juvenile steelhead trout and coho salmon (*Oncorhynchus kisutch*) were the most abundant species.

Measurement of territory size

We measured the territories of 50 fish, 25 in the Chilliwack River and 25 in the Salmon River, from 15 June to 30 September 1994. To measure territory size, we observed individual fish defending territories along the stream bottom from behind a portable blind set up on the stream bank. We recorded the aggressive interactions of territorial fish with intruding individuals by sketching a territory map onto a slate. To estimate a territory boundary, we recorded the aggressive distance of a fish under observation (hereafter the focal fish) to an intruding fish. Aggressive distance was defined as the distance between the focal fish and an intruder when the latter elicited an aggressive response from the focal fish. After observing each fish (observation time = 1 hr), actual measurements of the aggressive distances were made using a meter stick. In areas of the rivers where surface obstruction or depth precluded observing fish from behind the blind, we observed fish by snorkeling. Although fish are sensitive to overhead movement, approaching as a snorkeler from a downstream position, did not appear to disturb them because they continued to feed and interact with intruders. We used average aggressive distance as an estimate of territory radius and then calculated territory size as the area of a circle. We chose to use the 'circular' method rather than calculating territory area as the maximum convex polygon (*cf.* Schoener, 1981) because the circular method, unlike the polygon method, uses the mean radius that is not biased by the number of aggressive observations (Keeley & Grant, 1995).

After observing a fish, we attempted to capture it and were successful in 41 of 50 attempts. The fish were captured by angling or by using two aquarium dip nets. We then measured the fork length of individuals (Scott & Crossman, 1973) and preserved them in 10% formalin. We also visually estimated body length for each focal fish during an observation period and later compared the visually estimated body length with the actual body length of the 41 captured fish. The two length estimates did not differ significantly (paired $t = 1.028$, $p = 0.31$) and were on average only 5.8% different. In our statistical analyses, we used the actual length measurements for the 41 captured fish and we used visually estimated body lengths for the 9 uncaptured fish.

Measurement of environmental variables

After completing an observation period, we estimated food abundance by sampling invertebrates drifting through a fish's territory. We sampled drift by placing a drift net, 1 meter long with an opening of 15.2×23 cm and a mesh size of $300 \mu\text{m}$, on the center of a territory for 1 hour. The mouth of the drift net was constructed from a rectangular metal collar that was 12.5 cm deep. The function of the collar was to maintain laminar flow at the mouth of the net, and to help prevent backwashing (see Field-Dodgson, 1985). Each sample was placed in a 500 ml jar and later sorted to remove inedible material. After sorting, the remaining animals were then stored in 10% formalin until they could be measured using a microscope equipped with a digitizing computer system.

To eliminate drift items that were either too small or too large to be eaten by a focal fish, we used the stomach contents of the 41 captured fish to determine the maximum and minimum prey sizes that could be ingested by a fish of a given size. Average, maximum and minimum prey sizes were positively related to the fork length of the 41 captured fish (E.R. Keeley, unpubl. data). We used the upper 95% prediction interval from the maximum prey size regression and the lower 95% prediction interval from the minimum prey size

regression to eliminate any prey items in the drift sample that did not fall within the predicted ranges. Each drift sample was then dried in an oven at 50°C for 24 hours and moved to a desiccator for an additional 24 hours. Drift samples were then weighed at 24 hour intervals, until a stable weight (± 0.0002 g) was obtained. We used dry mass of invertebrates per m² of the water column profile per 1 hour drift sample, as our estimate of food abundance on a territory.

To estimate intruder pressure we counted the number of aggressive interactions between a focal fish and intruders, during each observation period. As a second measure, every 15 minutes, we counted local fish density surrounding the focal fish in a 1 and 4 m² area. Each of these measures of intruder pressure were used separately in our analyses. Following the definitions of Keenleyside & Yamamoto (1962), we classified the aggressive behaviour of territory holders towards intruders as being charges or lateral displays.

Statistical analyses

We combined measurements of each of these variables in an ordinary least squares multiple regression to determine their importance in predicting territory size. We used a type III or extra-sum-of-squares analysis to evaluate the effects of all independent variables and to ensure that no significant influence of multi-collinearity occurred (Neter *et al.*, 1990). For body size comparisons using mass as a size measure, we converted fish length measurements into mass with the following ordinary least-squares, length-mass regression for juvenile steelhead trout: $\log_{10} \text{ mass (g)} = 3.03 \log_{10} \text{ fork length (cm)} - 1.99$ (length range = 3.0 – 20.0 cm, $N = 3453$, $r^2 = 0.97$, $p < 0.0001$; B.R. Ward, B.C. Ministry of Environment, Lands and Parks, Vancouver, B.C., unpubl. data). Because body size measurements tend to be log-normally distributed, allometric equations were based on $\log_{10}$ transformed data (Harvey, 1982). We also tested additional variables for normality and $\log_{10}$ transformed them, if required.

Results

Frequency and spatial distribution of defence

Juvenile steelhead trout defend territories from a centrally located station from which they locate and expel intruding fish. The fish we observed initiated aggression from up to 5 different locations, but on average 80% of aggressive behaviour was from over the same location on the stream bottom. In 16% and 2% of cases, territory holders initiated aggression from a secondary or tertiary holding position, respectively. Of the 675 aggressive interactions we observed, 96% were directed towards conspecifics and the remaining 4% towards juvenile coho salmon. Eighty-nine percent of aggressive interactions were charges and 11% were lateral displays.

Ninety-six percent of aggressive behaviour consisted of charges towards conspecifics.

Aggressive distance did not differ whether a territory holder interacted with a conspecific or a heterospecific intruder (paired $t = 1.46$, $N = 9$, $p = 0.18$). Similarly, the distance at which a territory holder initiated aggression was not different if the individual used a lateral display or a charge to interact with an intruder (paired $t = 1.42$, $N = 25$, $p = 0.17$). Hence, we combined all estimates of aggressive distance to calculate territory size, whether the territory holder interacted with a conspecific or heterospecific, or whether it used a charge or lateral display. By comparing frequency and aggressive distance in body lengths by individual, we assessed the shape of territories defended by juvenile steelhead trout. We used Bonferroni-corrected (α -values with p -critical = 0.008, for all pair-wise comparisons (Neter *et al.*, 1990).

For comparative purposes, we categorized fish movement from holding stations into 12 equally divided directions. Territory holders defended space in each of the 12 directions, but the distribution of defence was not equally distributed among the categories (Fig. 1a; Rayleigh's test, $p < 0.001$; Batchelet, 1981). Despite differences in defence effort, territories appear to be bilaterally symmetrical because there was no significant difference between the left and right quadrants in terms of frequency of aggression (Fig. 1a; paired $t = 0.088$, $N = 39$, $p = 0.93$) or aggressive distance (Fig. 1b; paired $t = 1.54$, $N = 37$, $p = 0.13$). Similarly, frequency of aggression did not differ between the upstream quadrant and the average of the two lateral quadrants (Fig. 1a; paired $t = 0.88$, $N = 36$, $p = 0.39$) but aggressive distance was significantly greater upstream than laterally (Fig. 1a; paired $t = 3.91$, $N = 36$, $p = 0.0004$). In comparison to downstream areas, more defence effort was directed at the upstream quadrant (Fig. 1a; paired $t = 3.55$, $N = 37$, $p = 0.001$) and upstream aggressive acts also tended to be significantly longer (Fig. 1b; paired $t = 3.12$, $N = 37$, $p = 0.0035$). Aggressive distance downstream did not differ significantly from lateral quadrants (Fig. 1b; paired $t = 1.23$, $N = 32$, $p = 0.23$), but frequency of aggression was higher in comparison the downstream area (Fig. 1a; paired $t = 5.18$, $N = 32$, $p = 0.0001$).

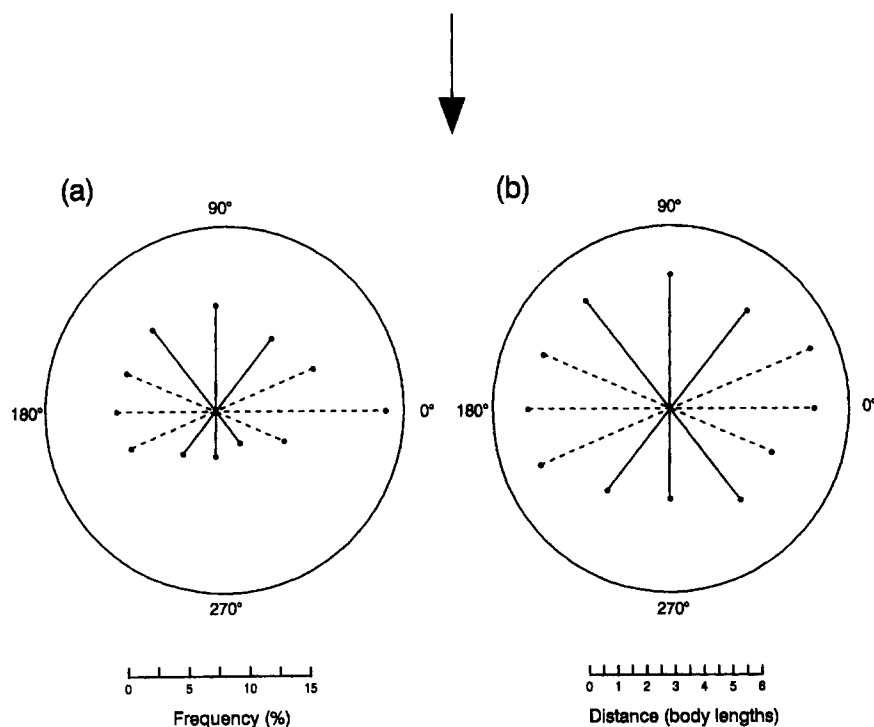


Fig. 1. (a) Relative frequency and (b) average distance, in body lengths, of aggressive interactions with intruders in 12 directions (675 acts by 50 focal fish). Dashed lines represent the lateral quadrants and the solid lines facing the top and bottom of the circles represent the upstream and downstream quadrants, respectively. For graphical purposes, all aggressive acts are assumed to be initiated from a single station. The arrow represents the direction of the current.

Correlates of territory size

As predicted, territory size of juvenile steelhead trout increased with increasing body length. The equation describing the simple allometric relationship is: $\log_{10} \text{territory size (m}^2\text{)} = 1.56 \log_{10} \text{fork length (cm)} - 1.81$ ($r^2 = 0.62$, $N = 50$, $p < 0.0001$). There also appeared to be two distinct groups of points within the allometric territory size relationship. In comparison to larger individuals, fish smaller than 5.0 cm appeared to have relatively large territories for their body size (Fig. 2a). We fit a piece-wise regression to describe this relationship (Neter *et al.*, 1990). There was no significant difference between the slopes of the two lines (ANCOVA, $F_{1,46} = 1.09$, $p = 0.30$), but their elevations differed significantly

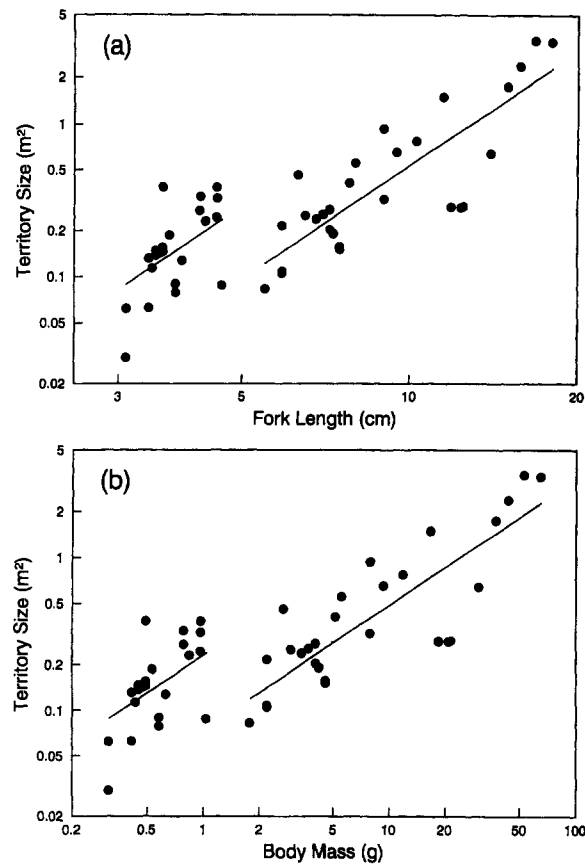


Fig. 2. The relationship between territory size and (a) fork length or (b) body mass for 50 juvenile steelhead trout.

(ANCOVA, $F_{1,46} = 13.78$, $p = 0.0005$). Hence, $\log_{10}$ territory size (m^2) = $2.47 \log_{10}$ fork length (cm) - 0.48 size category - 2.26 ($r^2 = 0.71$, $N = 50$, $p < 0.0001$), where size category is 0 for fish ≤ 5 cm or 1 for fish > 5 cm.

Territory size increased with body mass in a similar way, (Fig. 2b): $\log_{10}$ territory size (m^2) = $0.82 \log_{10}$ body mass (g) - 0.48 size category - 0.64 ($r^2 = 0.71$, $N = 50$, $p < 0.0001$). There was no significant difference between the slopes of the two lines (ANCOVA, $F_{1,46} = 1.09$, $p = 0.30$), but their elevations differed significantly (ANCOVA, $F_{1,46} = 13.78$, $p = 0.0005$).

In both the Chilliwack and Salmon rivers, the abundance of invertebrates drifting per m^2 of water column profile varied by an order of magnitude,

but showed overlapping ranges (Table 1). As expected, however, average abundance of drifting invertebrates was higher in the Salmon River in comparison to the Chilliwack River ($t = 4.22$, $p < 0.001$). In contrast, the local density of fish within each observation area did not differ between study streams ($t = 0.13$, $p = 0.90$). Observed rates of intrusion on territories were also not different between the two rivers ($t = 2.00$, $p > 0.05$).

We used a multiple regression approach to assess if food abundance and intruder pressure were significantly correlated with territory size. As predicted (Schoener, 1983), after controlling for the effects of all other variables entered in the model, territory size was inversely related to food abundance (Table 2). In contrast, the effects of intruder pressure were less clear. Territory size was not significantly related to intruder pressure when we used the number of intrusions observed on a territory per hour as a measure of defence costs (Table 2); however, after statistically controlling for the effects of body size and food abundance, territory size was significantly inversely related to local fish density (a second measure of intruder pressure; partial $r = -0.30$, $p = 0.038$).

From our observations, the multiple regression model that best describes territory size for juvenile steelhead trout is: $\log_{10}$ territory size (m^2) = $2.21 \log_{10}$ fork length (cm) $- 0.14 \log_{10}$ food abundance ($\text{mg dry mass}/\text{m}^2/\text{hr}$) $- 0.41$ size category $- 0.33$ fish density (number/m^2) $- 2.24$ ($r^2 = 0.74$, $p < 0.0001$), where size category is equal to 0 for fish ≤ 5.0 cm and 1 for fish > 5.0 cm.

Discussion

Juvenile steelhead trout defended feeding territories in the Salmon and Chilliwack rivers in a manner similar to that described for other stream-dwelling salmonids (Grant *et al.*, 1989; Elliott, 1990; Keeley & Grant, 1995). Defence effort was most frequently directed towards the upstream and lateral parts of the territories (Fig. 1a). Presumably, this gives territorial fish priority access to the cross-sectional area of the water column carrying invertebrate drift. Despite the concentration of effort in the upstream and lateral directions, steelhead trout also defended downstream areas that were not significantly different in size to lateral areas and were only approximately 25% smaller than the upstream quadrant (Fig. 1b). Although

TABLE 1. *Territory characteristics for juvenile steelhead trout in the Salmon and Chilliwack rivers, British Columbia*

Variable	Salmon River Mean	Salmon River Range	CV ^a	Chilliwack River Mean	Chilliwack River Range	CV ^a
Territory size (m ²)	0.25	0.063–0.64	55	0.71	0.03–3.42	140
Body size (fork length, cm)	7.24	3.4–14.0	39	6.83	3.1–18.0	72
Intruder pressure (intrusions/hr)	14.13	4–27	50	18.06	5–40	45
Food abundance (mg dry mass/m ² /hr)	216.33	18.02–993.05	101	68.98	2.86–222.25	77
Fish density (number/m ²)	2.86	1–7	50	2.71	1–5	39

^a Coefficient of variation (standard deviation/mean × 100).

TABLE 2. Multiple regression statistics for several independent variables predicting the territory size of juvenile steelhead trout

Variable	Coefficient	SS ^a	p-value ^a
Body size (fork length, cm)	2.37	3.86	0.0001
Size category (≤ 5 cm or > 5 cm)	-0.43	0.56	0.0027
Food abundance (mg dry mass/m ² /hr)	-0.19	0.26	0.034
Intruder pressure (intrusions/hr)	-0.003	0.014	0.62

^aSum of squares and probability for each variable when entered into the model after all other variables.

invertebrate drift captured by foraging fish originates upstream, exclusive use of downstream areas may also be important for capturing prey because fish are often displaced downstream while attempting to reach prey items in the current. The frequency and spatial distribution of defence effort we observed for juvenile steelhead trout is similar to that observed in both brook trout, *Salvelinus fontinalis* (Grant *et al.*, 1989) and Atlantic salmon, *Salmo salar* (Keeley & Grant, 1995).

Our data suggests that conspecifics make up the majority of intruders on steelhead trout territories, and although several other species of salmonids are present in both rivers, even abundant juvenile coho salmon were rarely observed interacting with steelhead trout. These observations support previous studies which have concluded that sympatric populations of salmonids, such as steelhead trout and coho salmon or coho and chinook salmon (*Oncorhynchus tshawytscha*), are spatially segregated within the same stream (Hartman, 1965; Taylor, 1991).

After statistically controlling for the effect of body size, increasing food abundance was negatively correlated with territory size for juvenile steelhead trout (Table 2). Theoretical models of optimal feeding territory size predict either an increase or decrease in territory size with increasing food abundance (Schoener, 1983). These contradictory predictions of territory size models (*e.g.* Ebersole, 1980; Hixon, 1980) occur because some assume time minimization as a goal, while others assume energy maximization (Schoener, 1983). Fish species are generally assumed to be energy maximizers because they have indeterminate growth and because increasing body size is usually correlated with fitness (Grant, 1997). Hence, if salmonid fishes are energy maximizers, models of feeding territory size

that are appropriate, will predict a decrease in territory size with increasing food abundance (Schoener, 1983).

Experimental studies of stream-dwelling salmonids that manipulated food abundance, have found an inverse relationship between food abundance and territory size in some cases (Slaney & Northcote, 1974), and not in others (Symons, 1971; McNicol & Noakes, 1984). Field studies have also provided conflicting results. Dill *et al.* (1981) found an inverse relationship between territory size of juvenile coho salmon and the abundance of benthic invertebrates, but not drifting invertebrates. Keeley & Grant (1995) found a weak, but significant relationship between invertebrate drift abundance and territory size after controlling for the effects of body size and other environmental variables. Our study of juvenile steelhead trout adds to the growing evidence that food abundance has a small inverse effect on territory size of salmonids, despite the fact that we measured territory size over a wide range of invertebrate abundance.

Optimal territory size models also predict that increasing costs of defence by increasing levels of intruders will decrease territory size for energy maximizers (Schoener, 1983; Grant, 1997). For some species of animals, such as birds (Myers *et al.*, 1979; Eberhard & Ewald, 1994) or coral reef fishes (Norman & Jones, 1984; Tricas, 1989), intruder pressure is the primary correlate of territory size. For stream-dwelling salmonids the effect is less clear. In two studies, both brook trout (McNicol & Noakes, 1984) and coho salmon (Dill *et al.*, 1981) defended smaller territories when population density or intruder pressure was higher; however, Keeley & Grant (1995) could not detect any significant relationship between territory size and intruder pressure for Atlantic salmon. In our study, we found a small but significant effect of local population density on territory size that accounted for a small proportion of the variance in territory size.

In previous studies of territoriality in stream-dwelling salmonids, body size has been shown to be a strong correlate of territory size; explaining from 70 to over 90% of the variation observed in territory size (Grant *et al.*, 1989; Elliott, 1990; Grant & Kramer, 1990; Keeley & Grant, 1995). The variability in the territory size of juvenile steelhead trout from the Salmon and Chilliwack rivers was also primarily explained by differences in body size (Fig. 2). Of the four previous studies that have examined the allometry of territory size in salmonids, only Atlantic salmon have

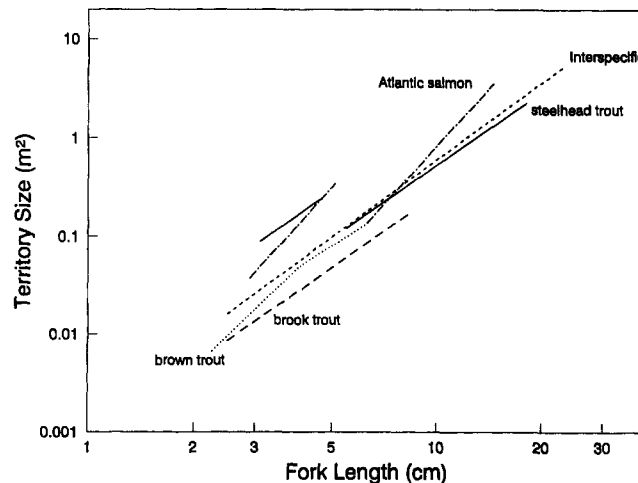


Fig. 3. The relationship between territory size and fork length for brook trout (dashed line; Grant *et al.*, 1989), brown trout (dotted line; Elliott, 1990); an interspecific study (short-dashed line; Grant & Kramer, 1990), Atlantic salmon (dashed-dotted lines; Keeley & Grant, 1995) and steelhead trout (solid lines; this study).

comparably sized territories (Fig. 3). The elevation of the steelhead trout and Atlantic salmon regressions were similar, twenty of the territory sizes we observed fell below the Atlantic salmon regression (Keeley & Grant, 1995) and 30 fell above (binomial test, $p > 0.2$). Over a similar range of fish sizes, all of our observations fell above regression for brook trout (Grant *et al.*, 1989) and 3 of 27 observations fell below the regression for brown trout (*Salmo trutta*; binomial test, $p < 0.001$; Elliott, 1990). Only 12 of 50 observations fell below Grant & Kramer's (1990) interspecific regression (binomial test, $p < 0.001$); however, this mainly due to the fact that steelhead trout < 5 cm in length always had larger territories than that predicted Grant & Kramer's (1990) equation, whereas fish > 5 cm fell above and below the line in nearly equal proportions. This suggests that only young-of-the-year steelhead trout were significantly larger than species other than Atlantic salmon.

There is a growing body of evidence suggesting that ontogenetic changes occur in the territory size defended by stream-dwelling salmonids. Elliott (1990) first described a change in the allometry of territory size for a salmonid when he found that juvenile brown trout larger than 4 cm, had territories that scaled to body size at a lower rate than fish smaller than

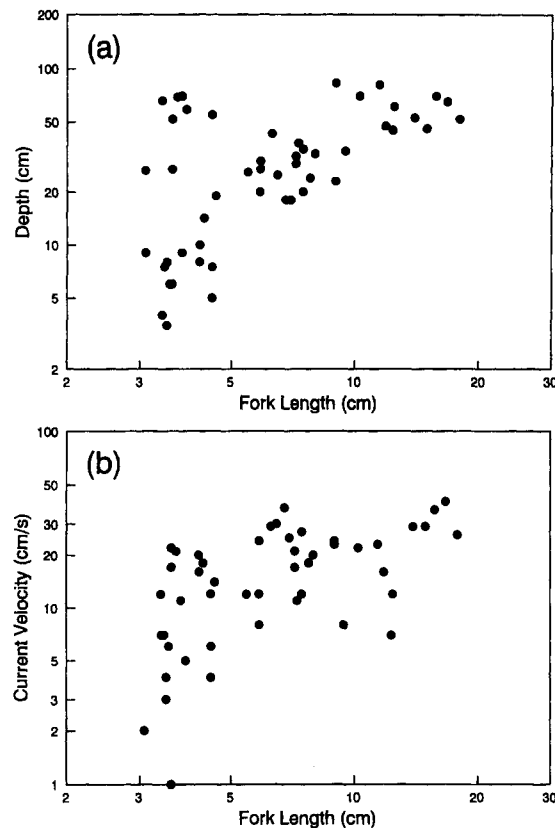


Fig. 4. The relationship between fork length of 50 juvenile steelhead trout and (a) depth ($r = 0.58$, $N = 50$, $p < 0.0001$) or (b) current velocity ($r = 0.59$, $N = 50$, $p < 0.0001$) of water flowing over their territories.

4 cm (Fig. 3). Keeley & Grant (1995) found that young-of-the year Atlantic salmon had relatively large territories for their body size in comparison to older fish, and we found a similar change in the regression elevation for steelhead trout (Fig. 2). Stream-dwelling salmonids are unique in comparison to other territorial animals in that they defend two-dimensional territories near the stream bottom, but they feed on invertebrate drift throughout the water column (Grant & Kramer, 1990). Because salmonids tend to select deeper and faster water as they grow (Everest & Chapman, 1972; Smith & Li, 1983; Keeley & Grant, 1995) and because deeper, faster water tends to carry more invertebrate drift (Britain & Eikeland, 1988), the productivity of a salmonid territory may depend on the cross-sectional profile

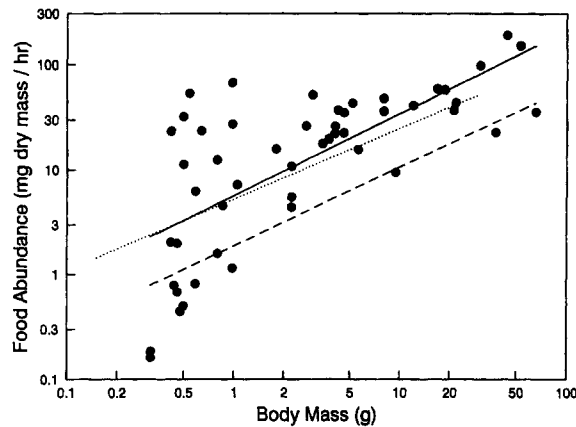


Fig. 5. The amount of invertebrate drift flowing through the territory of 50 juvenile steelhead trout per hour versus fish body mass ($\log_{10} Y = 0.83 \log_{10} X + 0.73$; $r^2 = 0.50$, $N = 50$, $p < 0.0001$; solid line). The dotted line represents the amount of food flowing through the territory of 48 juvenile Atlantic salmon (Keeley & Grant, 1995) and the dashed line represents the maximum ration per hour, predicted from Elliott's (1975) model for brown trout.

of water flowing over the area. Therefore, when fish move into deeper water as they grow, they may require relatively smaller two dimensional territories. As in other studies, the steelhead trout we observed tended to occupy deeper, faster water with increasing body size (Fig. 4a and b).

If distinct changes in territory size are a result of moving to more profitable areas of the stream as fish grow, then the amount of food captured by a salmonid territory should increase linearly with body size. Keeley & Grant (1995) assessed the relationship between the amount of food carried in the profile of water flowing through the territories of 46 juvenile Atlantic salmon and their body size. They found that the profile of food abundance increased linearly and scaled to body mass at a rate not different from that predicted for a similar sized brown trout fed on a maximum ration. We could also determine the mass of invertebrates flowing over the steelhead trout territories we observed according to fish size (Fig. 5); however, the variance in estimating invertebrate mass over territories, changed with fish size (Fig. 5). Hence, the relationship must be viewed cautiously. Nevertheless, the amount of food flowing through the profile of water over steelhead trout territories, increased with fish mass in a manner not significantly different from that of Atlantic salmon territories for

both slope (ANCOVA, $F_{1,94} = 1.33$, $p = 0.25$) and elevation (ANCOVA, $F_{1,94} = 0.001$, $p = 0.99$). Similarly, the slope was also not different from Elliott's (1975) maximum ration model for brown trout (Fig. 5; $t = 0.50$, $p > 0.5$). The elevation of this line was significantly greater than that predicted for a maximum ration (binomial test, $p < 0.01$); however, not all invertebrate drift that is available in a stream can be detected and eaten by fish (Wilzbach *et al.*, 1986). This may demand that territory holders defend space that captures more energy than would normally be required to meet energetic demands.

If territory size predicts the spatial requirements of salmonids in streams, then allometric territory size regressions may serve as useful predictors of the maximum densities of salmonids in streams as cohorts of fish increase in size (Grant & Kramer, 1990). Despite the strong relationship between territory size and body size, the variability in slope and elevation of the territory size equations (Fig. 4) suggests that there may be species or even population specific relationships. Therefore, if territory size relationships are useful in determining habitat saturation, based on space requirements (Grant & Kramer, 1990), then they may have to be applied to specific species or specific populations.

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